

BLUEPRINTING AZURE NETWORK TOPOLOGIES

BALANCING CONNECTIVITY, SCALE,
AND THE BLAST RADIUS



THE ASSUMPTION



THE REALITY



THE DEFAULT GAP

Azure VNets do not isolate workloads by default. The built-in AllowVNetInBound rule permits any resource to reach any other resource across all subnets.

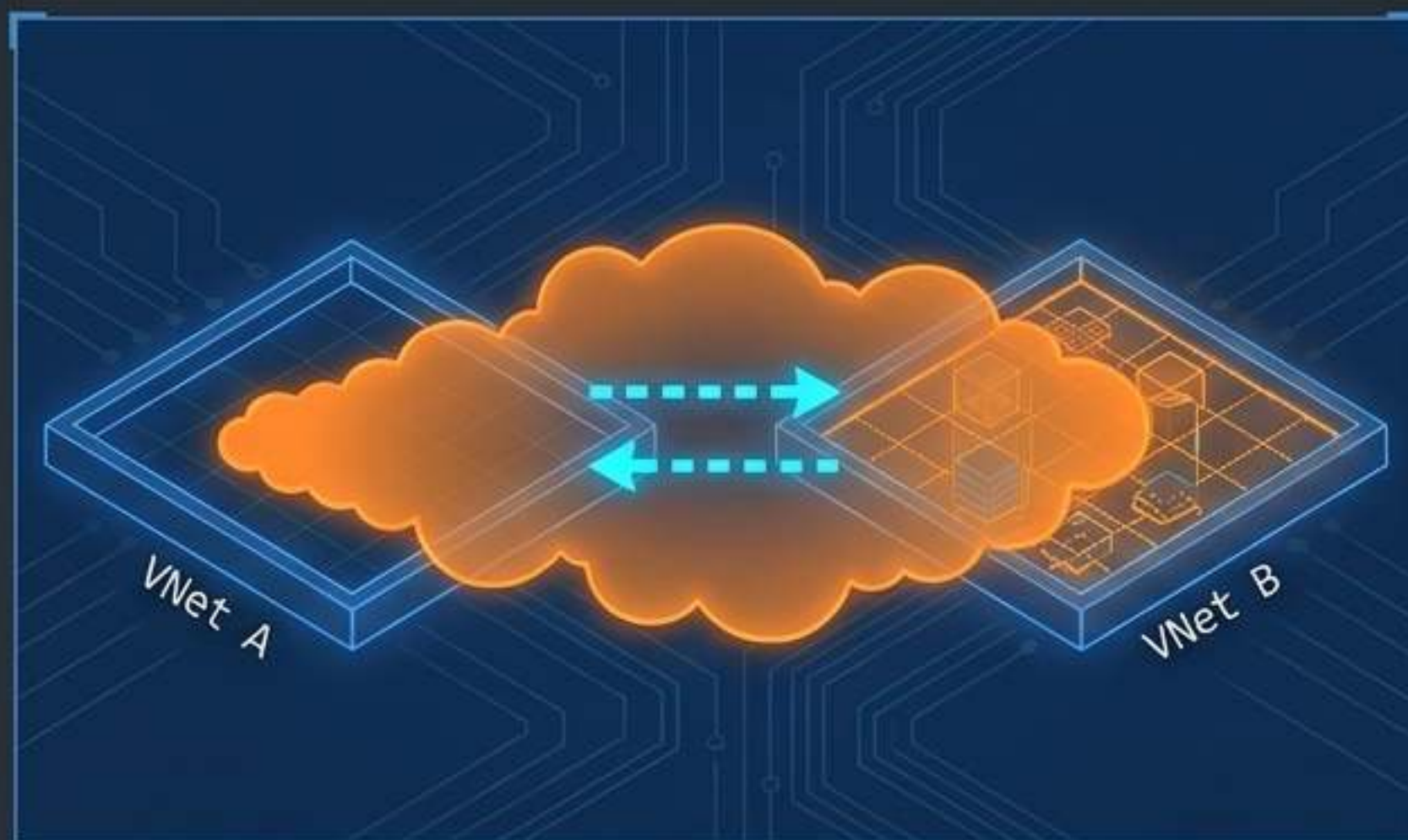


THE DATA PLANE THREAT

While Azure RBAC secures the management plane, unsegmented east-west traffic on the data plane enables unchecked lateral movement.



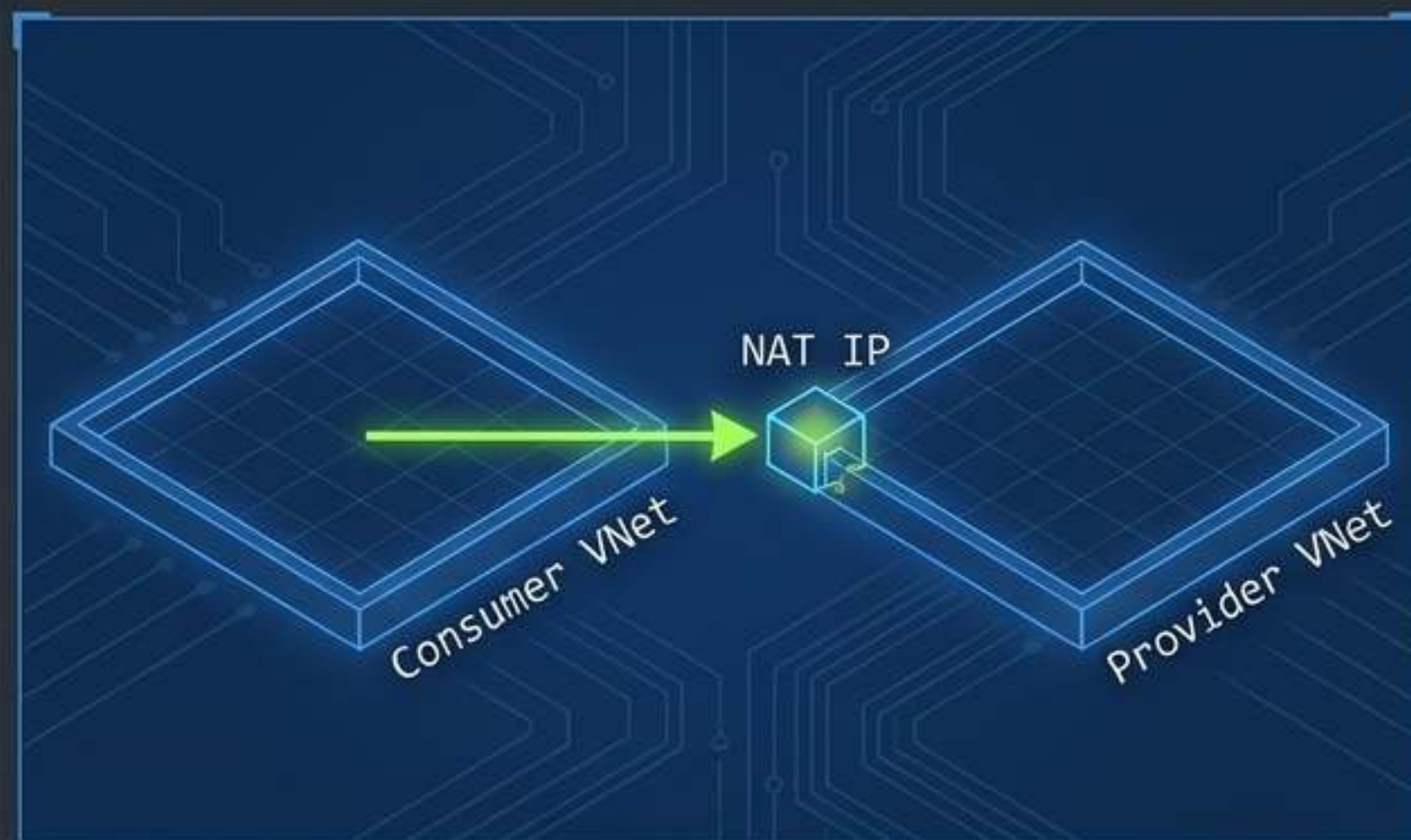
VNET PEERING



MECHANISM: Full, mutual network visibility. Subnets share routing natively.

BLAST RADIUS: High. A compromise in VNet A exposes all open ports in VNet B.

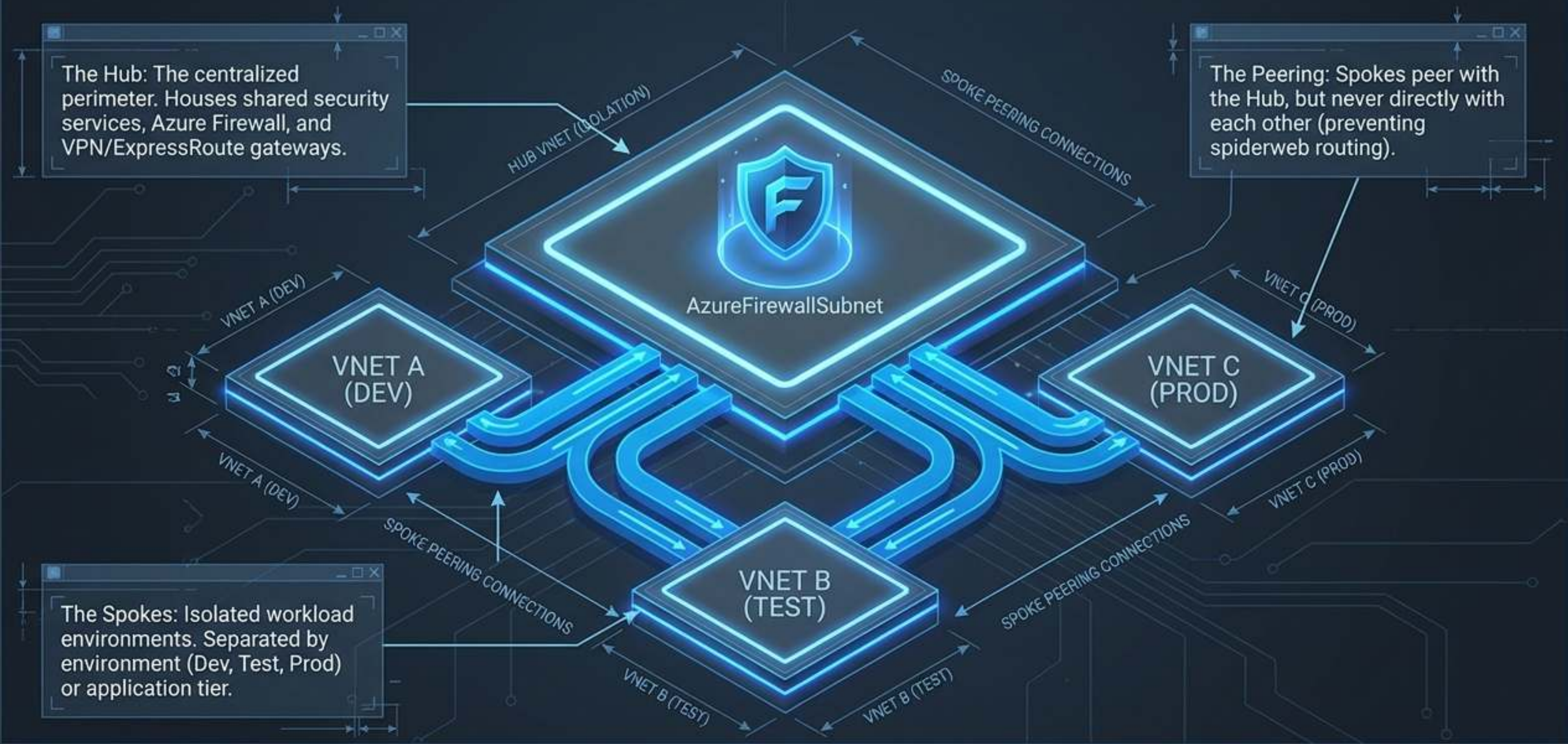
AZURE PRIVATE LINK SERVICE



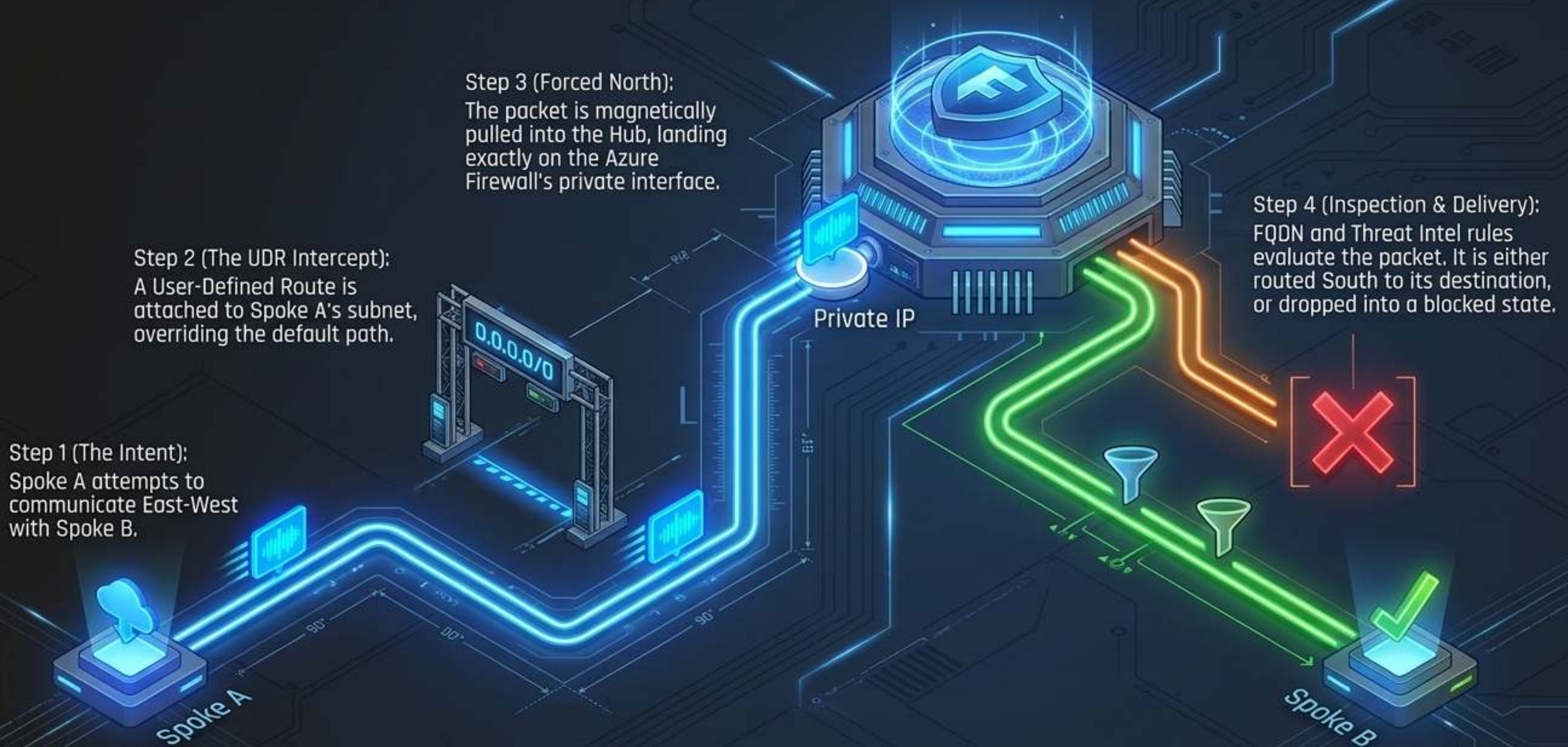
MECHANISM: Source NAT (SNAT) obscures the consumer IP. No overlapping IP conflicts.

BLAST RADIUS: Pinpoint. The consumer reaches exactly the exposed service and nothing else. No return initiation allowed.

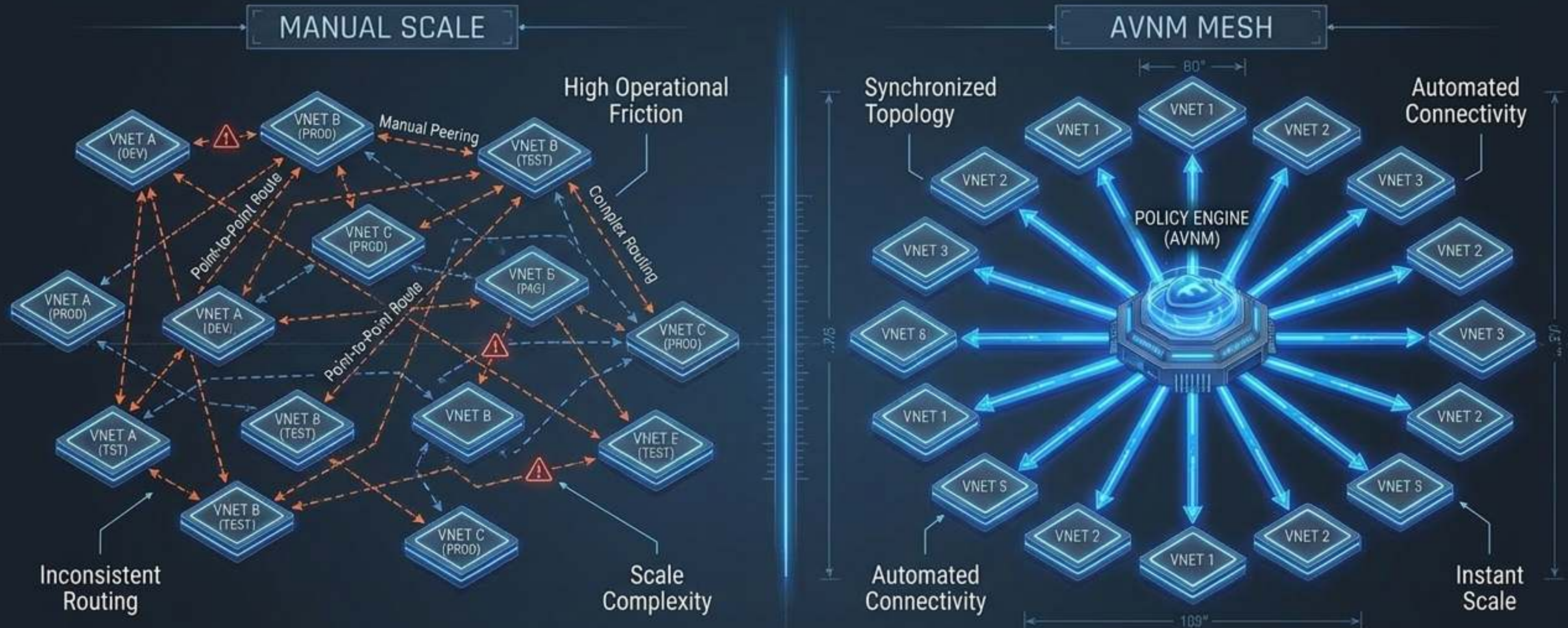
THE HUB & SPOKE ARCHITECTURE: CENTRALIZED SECURITY



DATA PACKET FLOW: CENTRALIZED SECURITY INTERCEPT & INSPECTION



NETWORK SCALE: MANUAL OVERHEAD VS. AVNM AUTOMATION



- **Dynamic Network Groups:** Logically grouping VNets based on Azure Policy conditions (e.g., environment:production).
- **Connectivity Configurations:** AVNM pushes standardized topology rules (Mesh or Hub-and-Spoke) instantly across multiple subscriptions, removing manual peering overhead.

NETWORK SECURITY RULE HIERARCHY & EVALUATION ORDER

LAYER 1: AVNM SECURITY ADMIN RULES

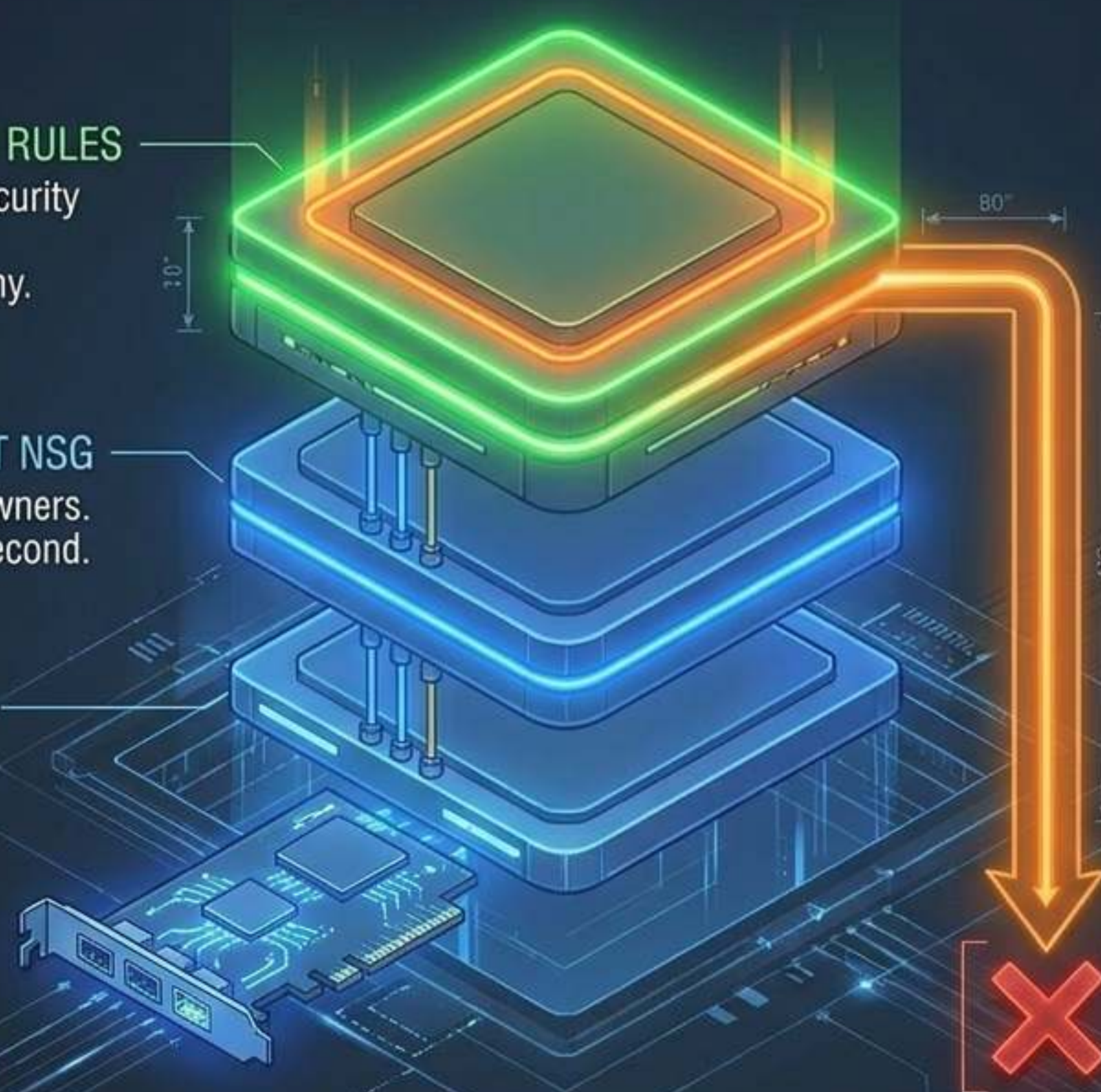
Managed exclusively by central security via AVNM. Evaluated first.
Supports Allow, Always Allow, Deny.

LAYER 2: SUBNET NSG

Managed by local workload owners. Evaluated second.

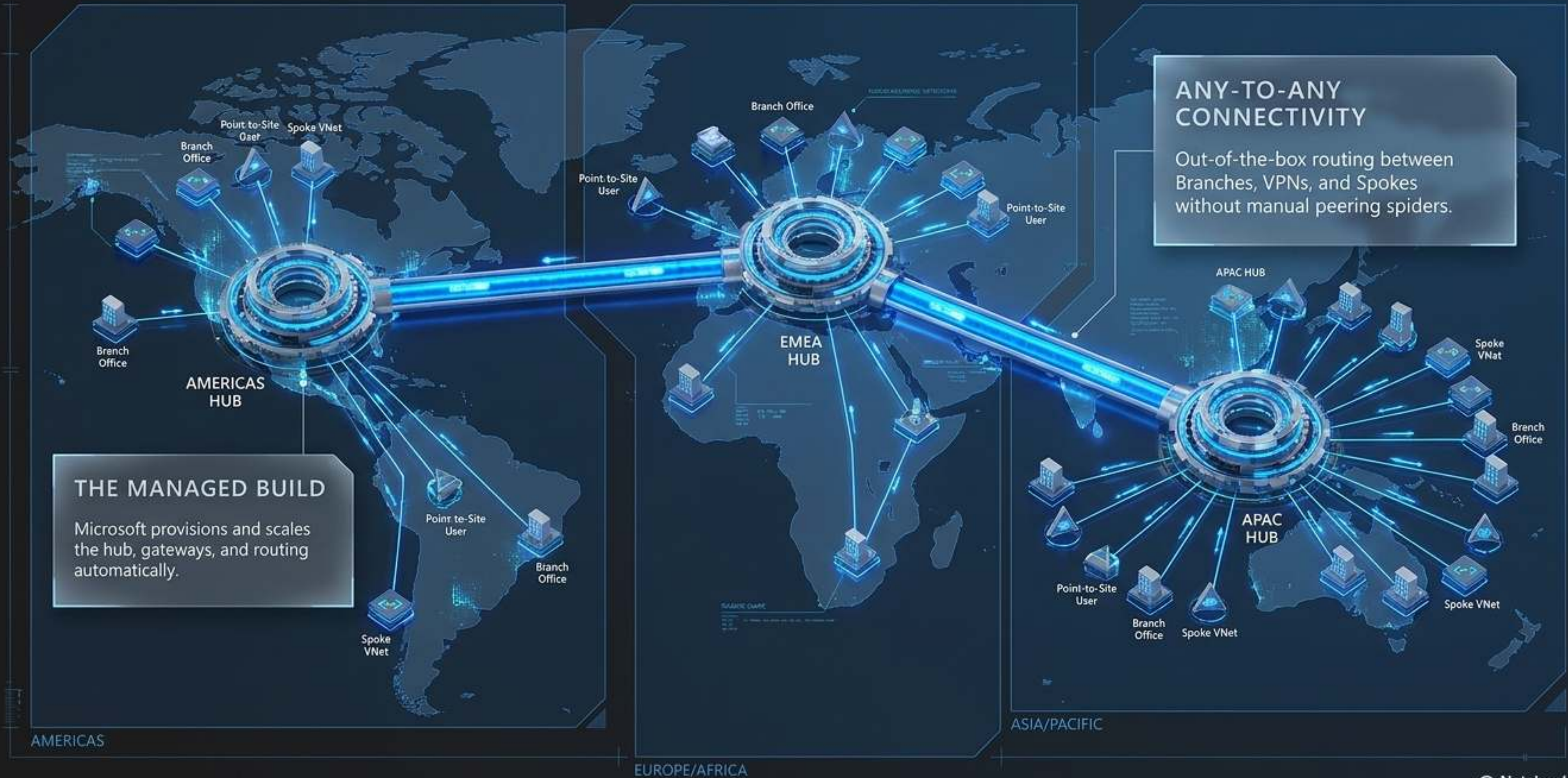
LAYER 3: NIC NSG

Attached to the specific VM interface. Evaluated last.



THE GOLDEN RULE: A centralized AVNM 'Deny 3389 (RDP)' rule blocks traffic globally, even if a local app owner explicitly allows it in their Subnet NSG.

AZURE VIRTUAL WAN: GLOBAL HUB-AND-SPOKE NETWORK



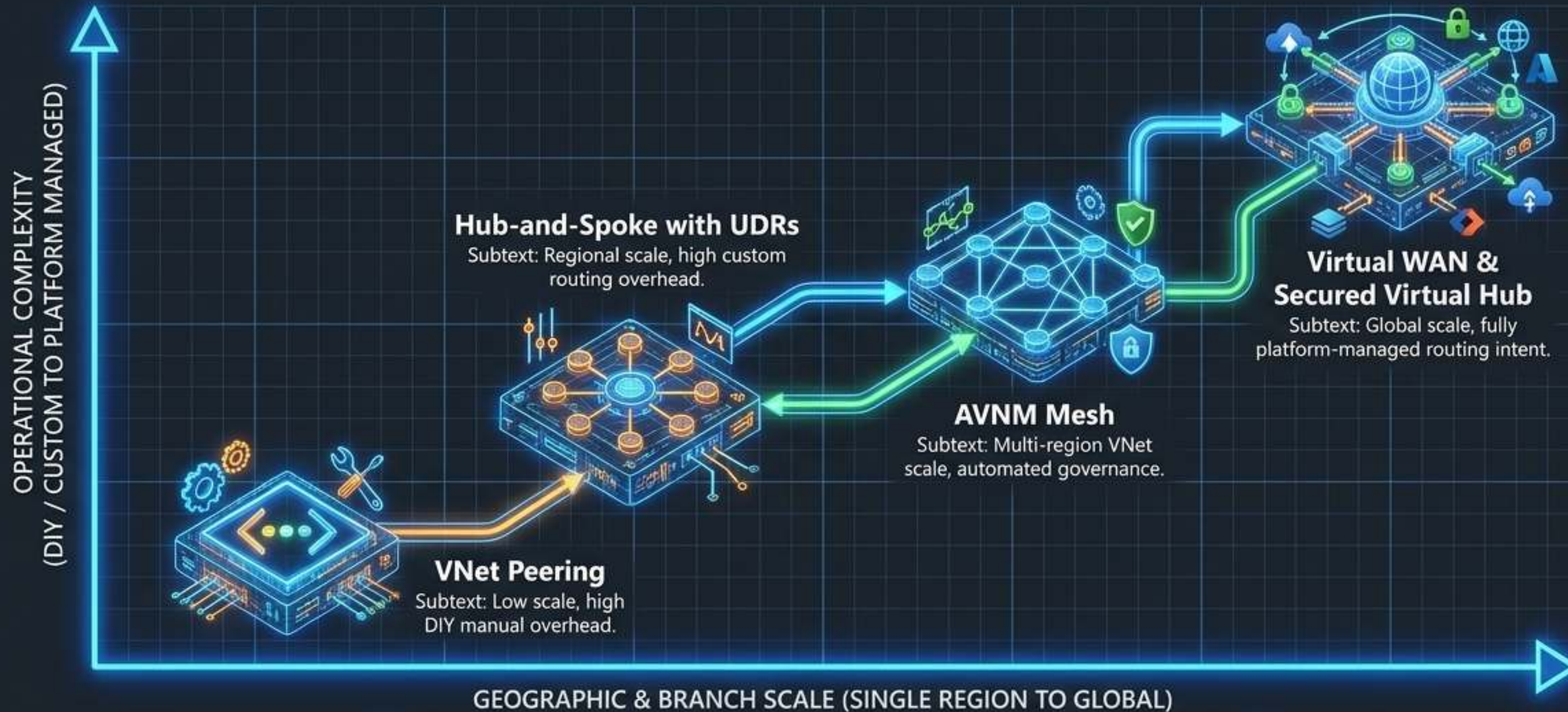
THE ROUTING INTENT FUNNEL



ARCHITECTURAL DECISION MATRIX: NETWORK HUB MODELS

	HUB-SPOKE VNET (CUSTOM)	SECURED VIRTUAL HUB (vWAN)
Branch (VPN/ExpressRoute) Connectivity	 Configured separately via individual gateways.	 Native, automated integration.
Route Management	 Manual UDRs mapped to each spoke.	 Automatic via Routing Intent.
Multi-Region Scale	 Requires manual peering and complex routing per region.	 Natively managed inter-hub routing on the Azure backbone.
Best-Fit Scenarios	 Single-region, VNet-only environments. High need for custom configurations.	 Multi-region, heavily branch-connected global enterprises.

THE ARCHITECTURAL JOURNEY: SCALING NETWORK COMPLEXITY



As your geographic scale and branch connectivity increase, shift the routing burden from the customer (Manual UDRs) to the platform (Routing Intent & AVNM).